

Corpus design and selection

Hakyung Sung
University of Hawaii at Manoa

September 9, 2026

Topics to cover

- Review
- Sampling and representativeness
- Different types of corpora
- Example study: Hanks et al. (2024)

Review: NLP in LCR

Overview of LCR

- Language data
- Methodological approaches
- Linguistic foci
- Contributions to SLA
- Contributions to L2 teaching

Overview of LCR

- Language data
 - Diverse L2 populations and language-use contexts need to be empirically investigated
 - Raw vs. annotated data
- Methodological approaches
- Linguistic foci
- Contributions to SLA
- Contributions to L2 teaching

Overview of LCR

- Language data
- Methodological approaches
 - CIA, CIA2
 - longitudinal, cross-sectional, pseudo-longitudinal, single corpus (descriptive)
- Linguistic foci
- Contributions to SLA
- Contributions to L2 teaching

Overview of LCR

- Language data
- Methodological approaches
- Linguistic foci
 - lexis, grammar, discourse, pragmatics...
 - written vs. spoken
- Contributions to SLA
- Contributions to L2 teaching

Overview of LCR

- Language data
- Methodological approaches
- Linguistic foci
- Contributions to SLA
 - descriptive
 - connect to SLA theory and test generalizability (e.g., universal order of acquisition of English morphemes)
 - connect to SLA theory and measure things (e.g., linguistic complexity)
 - connect to SLA theory and predict something (e.g., text readability)
- Contributions to L2 teaching

Overview of LCR

- Language data
- Methodological approaches
- Linguistic foci
- Contributions to SLA
- Contributions to L2 teaching
 - developing pedagogical materials
 - data-driven learning

Overview of LCR-NLP

- NLP pipeline
- tokenization
- lemmatization
- pos tagging
- parsing

NLP tool starts with input and output

At the most basic level,



- **Input:** What do I give the tool?
- **Output:** What exactly do I want back?
- **Processing:** What needs to happen in between?

Introduction

From research question to input

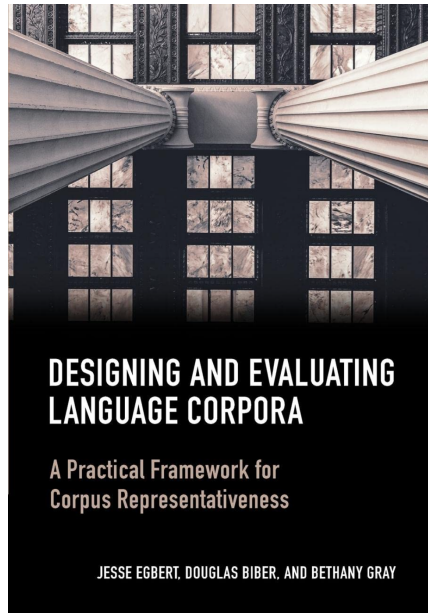
Suppose you are interested in a linguistic phenomenon X and want to investigate it empirically (i.e., using real examples of language use).

There are two basic ways to obtain your data:

- **Collect data**
- **Find existing data**

Q. Can we use just any collection of texts?
If not, what principles should guide corpus design?

Egbert et al. (2022), *Designing and Evaluating Language Corpora: A Practical Framework for Corpus Representativeness* provides one useful approach to answering this question.



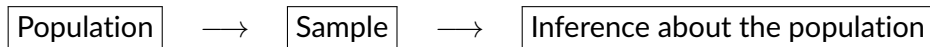
Sampling and representativeness

Population vs. sample

Suppose we want to understand the **reading habits of all university students**.

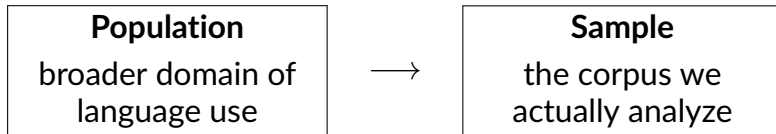
Can we ask every university student?

Usually, no. Instead:



- **Population:** the broader group we want to understand
- **Sample:** the smaller group we actually observe
- **Statistical inference:** using the sample to make careful generalizations about the population

Population vs. sample in corpus research



e.g.,

- **Population:** university academic writing
- **Sample:** 500 student essays collected from three universities

Population vs. sample in corpus research

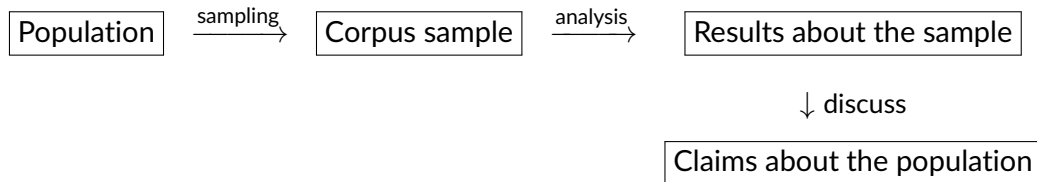
- **Population:** the broader group, domain, or type of language use that the researcher seeks to characterize.
- **Sample:** the actual collection of texts or language data selected from that population.

Typical examples of populations in LCR might include:

- academic writing produced by university students
- interaction in classrooms
- learner writing/speaking produced in standardized language tests
- argumentative/narrative essays written by second language learners
- ...

Why sampling matters

We observe the **sample**, but we often want to make claims about the broader **population**.



This brings us to representativeness

If a corpus is a **sample** of a broader population, we need to ask:

**How well does this sample represent
the population we want to understand?**

This is the basic question of **representativeness**.

A corpus should be designed/evaluated so that it can support **reasonable generalizations** about the language-use domain of interest.

Representative for what

“Different overall corpus designs represent different populations and meet different research purposes.”

Biber (1993, p. 245)

So we need to ask:

- **What population of language use** do I want to characterize?
- **What research purpose** should the corpus serve?

1. Size

There is no universal minimum size for a corpus.

Instead, we can ask:

- **How frequent is the feature I want to study?**
Rare linguistic features generally require ...
- **How much variation or comparison does my study need to capture?**
If the study includes multiple genres, tasks, speaker groups, proficiency levels, or other subgroups...

1. Size: Example

Suppose you have a corpus of 100 student essays.

Possibly sufficient

- average word length
- frequency of common function words
- mean sentence length

Possibly insufficient

- rare syntactic constructions
- low-frequency discourse features
- comparisons across many L1 groups

2. Principled sampling

Corpus design requires **explicit** decisions about:

texts · speakers/writers · tasks · genres · contexts

These sampling decisions should be motivated by:

Research question

+

Target domain

2. Principled sampling: Example

RQ. How does syntactic complexity differ between intermediate and advanced L2 English university writers in argumentative writing?

Study A

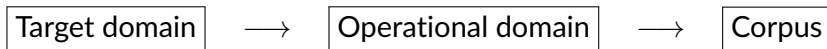
The corpus consisted of 500 essays written by university students learning English as a second language. The texts were collected from several university courses and represented learners at different proficiency levels. The corpus was used to examine syntactic complexity in learner writing.

Study B

The corpus consisted of 500 argumentative essays written by undergraduate L2 English students enrolled at three U.S. universities. The sample included 250 intermediate and 250 advanced writers, based on institutional placement scores. All texts were produced in response to one of four comparable argumentative prompts under a 40-minute timed-writing condition. Equal numbers of texts were sampled from the two proficiency groups at each institution.

3. Three-level framework

Before jumping directly from the population to the sample, we need an intermediate step. A useful distinction is between:



- **Target domain:** the broader language-use domain we ultimately want to characterize
- **Operational domain:** a concrete, researchable specification of that domain
- **Corpus:** the actual texts selected for analysis

3-1. From target domain to corpus

Suppose our target domain is:

L2 university academic writing

We must first specify what this means:

Operational domain

- Which universities?
- Which proficiency levels?
- Which genres or writing tasks?
- Which L1 backgrounds?
- Which time period?

Only then do we select the actual texts that constitute the **corpus**.

3-2. Two questions about representativeness

Evaluating representativeness involves two distinct questions:

1. Does the **operational domain** adequately capture the **target domain**?
2. Does the **corpus** adequately represent the **operational domain**?

Target domain $\xrightarrow{\text{conceptualization}}$ Operational domain $\xrightarrow{\text{sampling}}$ Corpus

3-3. Representativeness is relative

A corpus is not simply “representative” or “unrepresentative.”

3-3. Representativeness is relative: Example

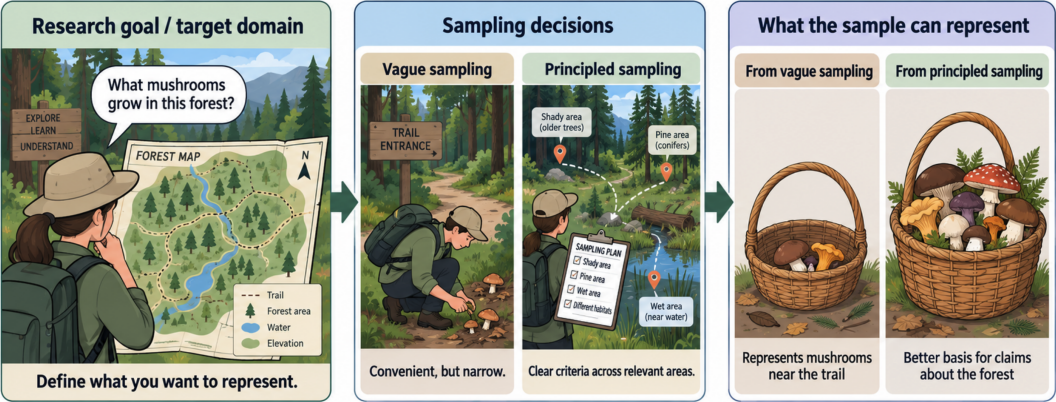


Image generated by ChatGPT

3-3. Representativeness is relative: Example

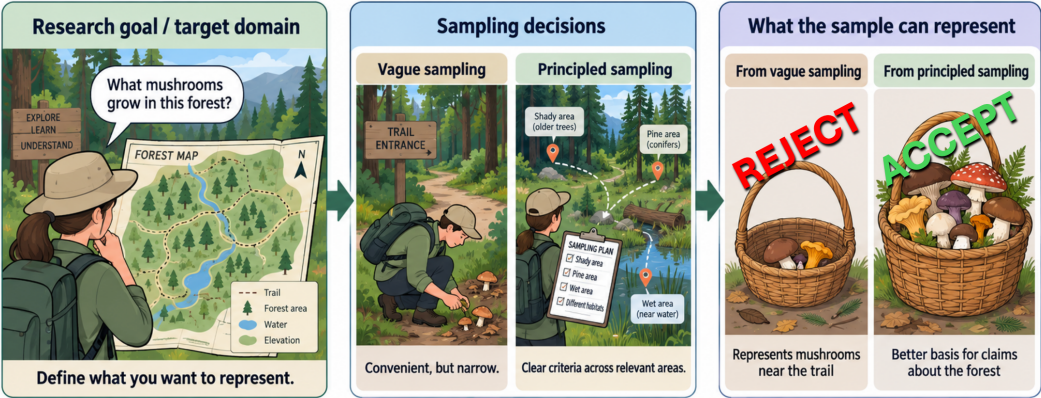


Image generated by ChatGPT + subject review

4. Evaluating representativeness

Two broad sets of considerations are important:

Domain considerations

- Is the target domain clearly defined?
- Does the operational domain capture it?
- Are relevant genres, tasks, speakers, contexts, or time periods included?

Distributional considerations

- Is the corpus large enough?
- Is there sufficient internal variation?
- Can it reliably estimate the linguistic variables of interest?

4-1. Confidence intervals (CIs)

Biber (1993) proposes using **confidence intervals (CIs)** to operationalize distributional representativeness.

4-1. Confidence intervals (CIs)

A corpus provides a sample from which we estimate a linguistic pattern.

For example, we might estimate, the mean sentence length across texts.

Corpus sample



Estimate of a linguistic pattern

For example:

Mean sentence length = 20 words

However, another sample from the same population would probably produce a slightly different estimate.

4-2. Sampling uncertainty

Imagine repeatedly drawing different samples from the same population.

We might obtain:

19.6 20.3 19.9 20.5 19.8

The estimate changes slightly from sample to sample (i.e., sampling uncertainty).

A confidence interval provides a way to express this uncertainty around our estimate.

4-3. What is a CI

A CI is a range around a sample-based estimate that reflects sampling uncertainty.

$$\boxed{\text{Estimate}} \pm \boxed{\text{Margin of error}}$$

e.g.,

$$20 \pm 1$$

corresponds to:

$$\boxed{19 \text{ to } 21}$$

The estimate is 20, but the interval indicates how precise that estimate is.

4-4. Interpretation of CI

Narrower CI $\Rightarrow$ More precise estimate

Wider CI $\Rightarrow$ Less precise estimate

For example:

20 [19.8, 20.2] vs. 20 [17, 23]

Both have the same estimated mean, but the first estimate is more precise.

4-5. More about CI

Two considerations are important:

- **Sample size:** More observations generally reduce sampling uncertainty and produce a narrower CI.
- **Variation across texts** Greater variation in the linguistic feature generally produces a wider CI.

More data + less variation



Narrower CI

Less data + more variation



Wider CI

cf. Why does sample size matter?

For the **sample mean**, the standard error is:

$$SE = \frac{s}{\sqrt{N}}$$

s = sample standard deviation N = sample size

The confidence interval is constructed using the standard error:

$$95\% \text{ CI} \approx \text{Sample mean} \pm 1.96 \times SE$$

Therefore:

$$\boxed{N \uparrow} \implies \boxed{SE \downarrow} \implies \boxed{\text{Narrower CI}}$$

4-6. From CIs to corpus design

We can now connect confidence intervals to **distributional representativeness**.

Q. Is our corpus large enough to estimate the linguistic feature of interest with sufficient precision?

This depends on:

- the *size* of the corpus;
- the *frequency* of the linguistic feature; and
- the *variation* in that feature across texts.

4-6. Setting a target level of precision

To evaluate whether a corpus is sufficiently large, we first need to decide how much sampling error we are willing to tolerate. One practical benchmark is:

$$\text{Target CI margin} = \pm 5\% \text{ of the observed mean}$$

Suppose the observed mean is:

$$100$$

Then:

$$5\% \text{ of } 100 = 5$$

and the desired margin is:

$$100 \pm 5$$

4-6. Setting a target level of precision

Once we define the desired level of precision, we can ask:

How many texts are needed to achieve it?

Frequent / less variable feature $\longrightarrow$ Smaller required N

Rare / more variable feature $\longrightarrow$ Larger required N

4-7. How large should the sample be?

Suppose we want to estimate the mean frequency of a linguistic feature across texts using a 95% CI. For the sample mean:

$$95\% \text{ CI} \approx \text{Sample mean} \pm 1.96 \times \frac{s}{\sqrt{N}}$$

The quantity added to and subtracted from the sample mean is the **margin of error**:

$$E = 1.96 \times \frac{s}{\sqrt{N}}$$

4-7. How large should the sample be?

If we want the margin of error to be no more than $\pm 5\%$ of the observed mean:

$$E = 0.05 \times \text{Mean}$$

We can calculate the required sample size:

$$N = \left(\frac{1.96s}{E} \right)^2$$

4-7. How large should the sample be?

Suppose a linguistic feature has:

$$\text{Mean} = 100 \quad s = 10$$

We want a margin of error of $\pm 5\%$ of the mean.

Step 1. Calculate the desired margin of error.

$$E = 0.05 \times 100 = ?$$

$$E = 5$$

Step 2. Calculate the required sample size.

$$N = \left(\frac{1.96 \times 10}{5} \right)^2$$

What is the required N ?

4-7. How large should the sample be?

Feature	Mean	s	Required N
Feature A	100	20	?
Feature B	100	40	?
Feature C	100	80	?

For each feature:

$$E = 0.05 \times 100 = 5$$

$$N = \left(\frac{1.96s}{5} \right)^2$$

Discuss:

1. Which feature requires the largest corpus?
2. What happens to required N as variability increases?
3. What does this tell us about distributional representativeness?

Different types of corpora

Different types of corpora

Different research questions require different kinds of corpora (related to *representativeness* and *balance* to the corpus design framework; Gries, 2009).

- General vs. specific
- Diachronic vs. synchronic
- Monolingual vs. parallel
- Static vs. dynamic

What information can a corpus contain?

A corpus may contain more than the language data itself.

- Language data
- Metadata: information about the data
- Markup: information about document structure
- Annotation: linguistic analysis added to the data

Metadata $\neq$ Markup $\neq$ Annotation

Different ways to process corpora

Researchers need ways to retrieve, transform, annotate, and analyze corpus data.

Corpus analysis programs

- AntConc
- WordSmith Tools
- #LancsBox
- Sketch Engine

Typical functions:

- concordances
- frequency lists
- keywords
- collocations
- n-grams

Programming (e.g., Python, R)

Can be used to:

- clean/restructure corpus files
- customize searches
- extract annotated features
- normalize frequencies
- merge metadata
- visualize and statistically analyze results

Example study: Hanks et al. (2024)